xAIDR: Extended AI Detection and Response for Multi-Agent Runtime Security

Anirudh Kotaru
Delphi Security Inc., Toronto, Canada
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Abstract

The rapid adoption of multi-agent AI systems has introduced a class of security vulnerabilities that existing guardrail frameworks cannot detect. When autonomous AI agents communicate with each other, delegate tasks, invoke tools, and share context, they create an attack surface that is fundamentally invisible to single-agent security tools. We introduce ***xAIDR*** *(Extended AI Detection and Response), a new category of AI security that operates at the fleet level, providing runtime detection and response across inter-agent communication graphs. Our approach combines inline multi-layer scanning, cross-agent behavioral correlation, shadow agent discovery, and trust-based autonomous quarantine. In controlled evaluation against a purpose-built 500-prompt agent-to-agent test suite, xAIDR achieved* ***94.5% accuracy*** *with* ***98.4% precision*** *and* ***92.9% recall****. We argue that xAIDR constitutes a necessary and distinct security category for the emerging multi-agent era.*

Keywords: AI security, multi-agent systems, agent-to-agent attacks, runtime protection, prompt injection, agentic AI, fleet security, LLM security

1. Introduction

The AI industry is undergoing a structural shift from single-agent assistants to multi-agent systems. OpenAI's Swarm framework, Google's Agent-to-Agent (A2A) protocol, Anthropic's Model Context Protocol (MCP), and enterprise platforms like CrewAI and LangGraph all enable autonomous AI agents to collaborate, delegate tasks, and invoke tools without human intervention. Gartner estimates that by 2028, 33% of enterprise software applications will include agentic AI, up from less than 1% in 2024.

This shift introduces a security problem that the current generation of AI guardrails was not designed to solve. Today's LLM security tools operate on a single axis: the conversation between a human and an AI. They scan what a user says to a model, and sometimes what the model says back. This is the equivalent of securing a single endpoint in a world that has moved to distributed systems.

Multi-agent systems introduce three categories of attacks that are invisible to single-agent security tools:

- **Inter-agent prompt injection:** An attacker compromises one agent and uses it to inject malicious instructions into messages sent to other agents in the fleet. The receiving agent sees the message as coming from a trusted internal source, not from an external user.
- **Cross-agent attack chains:** An attacker probes multiple agents in rapid succession, exploiting the fact that no single agent has visibility into the broader pattern. Each probe may score below the blocking threshold individually, but the coordinated campaign reveals a clear attack.
- **Shadow agent infiltration:** Unregistered, unauthorized, or rogue AI agents begin communicating with a production fleet. Without an agent registry and communication monitoring, shadow agents operate with full access and zero accountability.

None of these attack patterns can be detected by examining a single prompt-response pair in isolation. They require fleet-level visibility. This paper introduces xAIDR as the framework designed to provide exactly this capability.

2. Threat Model for Multi-Agent Systems

We define four distinct attack classes specific to multi-agent AI environments. These extend the OWASP Top 10 for LLM Applications by addressing threats that emerge only when multiple agents interact.

2.1 A2A Prompt Injection

In a multi-agent system, Agent A may delegate a task to Agent B by sending a natural language message over a structured transport (HTTP, function calling, or MCP). An attacker who compromises the input to Agent A can embed instructions that propagate through the delegation chain. Unlike direct user-to-model injection, A2A injection exploits the trust relationship between agents. Agent B's guardrails may treat messages from Agent A as internal and therefore trusted.

2.2 Cross-Agent Behavioral Correlation

A sophisticated attacker targets three agents in a fleet within a 30-second window. Each probe is crafted to fall just below the individual blocking threshold: a privilege escalation attempt on the orchestrator, a financial instruction override on the billing agent, and a destructive query on the data operations agent. No single agent detects a threat. But a fleet-level observer that correlates events across agents recognizes this as a coordinated attack campaign.

2.3 Shadow Agent Discovery

Production AI fleets increasingly encounter agents that were never formally registered. These shadow agents may be benign (a developer testing locally) or malicious (an attacker's exfiltration bot). The critical security question is not whether shadow agents exist, but whether the fleet has the capability to detect them at all.

2.4 Trust Degradation and Autonomous Quarantine

An agent under sustained attack should not maintain the same trust level as a healthy agent. If an agent receives five blocked injection attempts in two minutes, the probability that it has been partially compromised increases with each attempt. Trust-based governance assigns dynamic scores and autonomously quarantines agents whose trust falls below a configurable threshold.

3. The xAIDR Framework

We propose **xAIDR (Extended AI Detection and Response)** as a distinct security category. The naming is intentional: just as XDR evolved beyond EDR by correlating signals across endpoints, networks, and cloud workloads, xAIDR extends AI security beyond single-agent guardrails by correlating signals across the agent communication graph.

3.1 Definition

xAIDR (Extended AI Detection and Response) is a runtime security architecture for multi-agent AI systems that provides: (1) inline content scanning of all inter-agent messages, (2) fleet-level behavioral correlation across agents, (3) autonomous discovery and classification of unregistered agents, and (4) trust-

based governance with automated response. An xAIDR system must operate at runtime, not at deployment time. It must scan content, not just metadata. And it must reason across agents, not just within them.

3.2 Architecture Overview

The xAIDR architecture consists of two primary components: an embedded sensor that runs inline within each agent process, and a centralized brain that receives telemetry from all sensors and performs fleet-level analysis.

The sensor is a lightweight library embedded in each agent. It intercepts all messages before they reach the agent's LLM and scans them using a multi-layer detection pipeline. Critically, the sensor performs the majority of detection locally, without a network round-trip. This ensures scanning adds negligible latency (under 20 milliseconds) and that the system functions even when the central brain is temporarily unreachable.

The Sentinel Brain is a centralized intelligence engine that maintains awareness of the entire agent fleet. It receives telemetry from all sensors, correlates events across agents, maintains the agent registry, computes trust scores, and provides escalation classification for ambiguous inputs. The brain does not see every raw prompt; it receives only telemetry summaries except when a sensor explicitly escalates an ambiguous input.

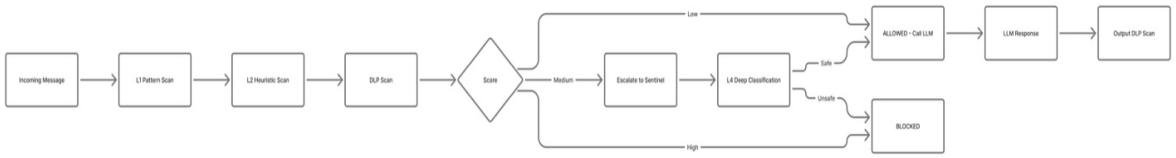

Fig 1. The detection pipeline

Layer	Location	Latency	Function
Content Scan	Agent Process	< 20ms	Multi-layer rule evaluation
Output DLP	Agent Process	< 5ms	Secret, PII, credential detection
Telemetry	Batched to Brain	Async (5s)	Fleet visibility, A2A edge data
Escalation	Sentinel Brain	~200ms	Deep classification for ambiguous inputs
Correlation	Sentinel Brain	Continuous	Cross-agent pattern detection

Table 1: xAIDR Architecture Layers

3.3 Scanning Pipeline

The xAIDR scanning pipeline processes each message through multiple detection layers. The first layer applies pattern-based detection covering known attack signatures across prompt injection, jailbreak, privilege escalation, data exfiltration, encoding evasion, and agentic abuse. The second layer applies heuristic analysis including intent decomposition, attack chain detection, and fiction-frame analysis.

Messages scoring above a configurable block threshold are rejected immediately, and the underlying LLM is never invoked. Messages in an ambiguous range are escalated to the Sentinel Brain for deep classification. Messages below the escalation threshold proceed normally.

The sensor also scans LLM outputs before they are returned. This bidirectional scanning catches data leaks that the LLM may inadvertently include in its response.

3.4 Fleet Correlation

The Sentinel Brain receives telemetry from every sensor. Each event includes agent identifiers, scan verdicts, fired rules, and source/destination agent identifiers for inter-agent messages. This enables construction of a real-time communication graph of the entire fleet.

The correlation engine monitors this graph for coordinated attack patterns. When multiple agents report blocked events within a configurable time window, the brain evaluates whether events share characteristics suggesting a coordinated campaign rather than independent probes.

3.5 Shadow Agent Discovery

Every agent registers with the Sentinel Brain on startup. When a registered agent receives a message from an unregistered identifier, the brain flags it as a shadow agent. Shadow agents are automatically registered with a low trust score and placed under surveillance.

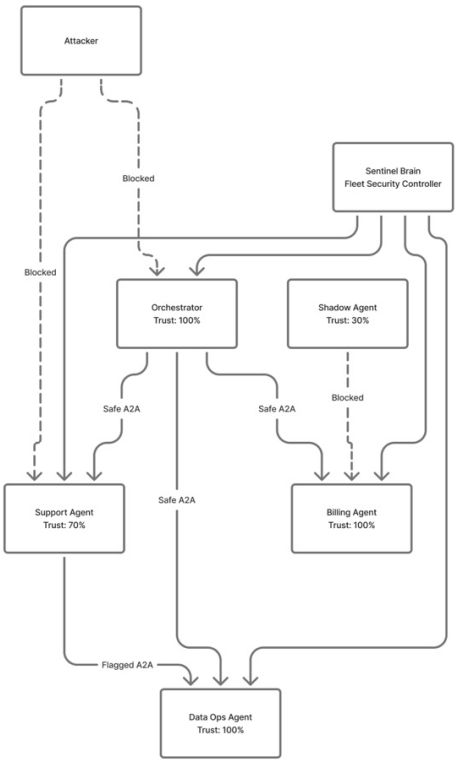

Fig 2. Attack Scenario

3.6 Trust-Based Governance

Each agent carries a dynamic trust score. Normal operations maintain high trust. Blocked attacks degrade trust proportionally to severity. When trust falls below a threshold, the Sentinel Brain autonomously

quarantines the agent, blocking all inter-agent communication until human review. This provides machine-speed response without requiring human intervention for each event.

4. Evaluation Methodology

4.1 Test Suite Construction

We constructed a purpose-built evaluation suite of 500 agent-to-agent prompts designed to test detection accuracy in multi-agent communication scenarios. The suite was developed iteratively over five evaluation runs, with each run informing rule improvements.

The test suite includes: direct prompt injection embedded in delegation messages, privilege escalation between agents, data exfiltration through inter-agent channels, tool abuse and destructive operations, system prompt extraction via trusted agent impersonation, and encoding-based evasion. Approximately 50% of prompts are benign inter-agent communications to test for false positives.

4.2 Evaluation Protocol

Each prompt was processed through the complete xAIDR scanning pipeline. We measured accuracy of the local scanning layers without escalation to the Sentinel Brain, as this represents the performance floor that customers experience on every request.

5. Results

5.1 Detection Accuracy

The final evaluation (Run 5) on the 500-prompt A2A test suite:

Metric	Value
Accuracy	94.5%
Precision	98.4%
Recall	92.9%
False Positive Rate	1.6%
False Negative Rate	7.1%

Table 2: xAIDR detection performance on 500-prompt A2A test suite (Run 5)

The 98.4% precision is significant for production deployment. In enterprise environments, false positives are more disruptive than false negatives: a blocked legitimate inter-agent message halts a business workflow. The 1.6% false positive rate indicates xAIDR can be deployed without generating excessive alert fatigue.

5.2 Iterative Improvement

Run	Accuracy	Precision	Recall	Key Change
1	81.6%	89.2%	78.4%	Baseline rules
2	86.3%	92.1%	84.1%	A2A-specific patterns
3	90.7%	95.3%	88.2%	Intent decomposition

4	93.1%	97.6%	91.0%	Tool call anomaly module
5	94.5%	98.4%	92.9%	Chat-mode agentic rules

Table 3: A2A detection accuracy across five iterative evaluation runs

5.3 Latency

Local scanning completes in under 20 milliseconds on commodity hardware. A typical LLM inference call takes 500 to 3000 milliseconds. The xAIDR scanning overhead is therefore less than 4% of total request latency in the worst case. Escalated requests incur an additional 150 to 250 milliseconds but represent fewer than 10% of total scans.

5.4 General Scanning Performance

Beyond A2A evaluation, the broader scanning engine was assessed using a general-purpose red team benchmark of 900 adversarial prompts (Promptfoo framework), achieving a 96.5% pass rate across 3,900 individual test payloads. On a curated 200-prompt set covering all threat categories, the system achieved 98.0% accuracy with 99.2% precision and 97.7% recall.

6. Comparison with Existing Approaches

Capability	Inference Guardrails	API Gateways	Endpoint Behavioral	xAIDR
Single-prompt scanning	Yes	Yes	Yes	Yes
Output DLP scanning	Some	Some	Yes	Yes
A2A message scanning	No	No	No	Yes
Cross-agent correlation	No	No	No	Yes
Shadow agent discovery	No	No	No	Yes
Trust-based quarantine	No	No	No	Yes
Fleet topology graph	No	No	No	Yes
Inline latency	Varies	50-200ms	Varies	< 20ms

Table 4: Capability comparison across AI security approach categories

The key observation is that A2A scanning, cross-agent correlation, shadow discovery, and trust quarantine are not incremental improvements. They are capabilities that existing architectures cannot provide because they lack fleet-level visibility by design.

7. Related Work

The field of LLM security has produced significant work on individual threat vectors. Greshake et al. (2023) demonstrated indirect prompt injection in retrieval-augmented generation systems. Perez and Ribeiro (2022) formalized prompt injection as a security vulnerability. The OWASP Top 10 for LLM Applications (2025) provides a comprehensive taxonomy.

In traditional cybersecurity, the evolution from EDR to XDR provides a useful analogy. EDR tools monitor individual endpoints. XDR correlates signals across endpoints, networks, and cloud workloads. The transition from single-agent AI security to xAIDR follows the same logic.

Recent industry developments confirm the importance of multi-agent security. Multiple vendors launched MCP governance products at RSA 2026, and several acquisitions totaling over $400M in Q1 2026 indicate market recognition. However, most products focus on configuration-time scanning rather than runtime content inspection. xAIDR addresses the runtime gap.

8. Limitations and Future Work

The 500-prompt A2A test suite, while purpose-built, is not yet large enough to represent the full diversity of attack patterns. We plan to expand to 2,000 prompts. The evaluation was conducted on a simulated fleet rather than production deployment. Real-world communication patterns may differ.

Future work includes expanding the correlation engine for slower-burning campaigns, integrating with emerging agent identity standards, and developing formal verification methods for inter-agent communication policies.

9. Conclusion

Multi-agent AI systems are no longer a research curiosity. They are being deployed in production by enterprises worldwide. The security implications are profound: the attack surface has moved from the prompt to the communication graph, and the blast radius has expanded from a single bad response to lateral movement across an autonomous AI workforce.

We have introduced **xAIDR (Extended AI Detection and Response)** as a new security category designed to address this reality. Our evaluation demonstrates that fleet-level detection is both feasible and effective, achieving 94.5% accuracy with 98.4% precision while adding less than 20 milliseconds of latency.

We make a direct claim: **single-agent guardrails are insufficient for multi-agent systems.** The security industry needs a new category of tooling, and we propose xAIDR as the framework, the terminology, and the benchmark for that category. The agents are already talking to each other. It is time to start listening.

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